Finance, Firm Dynamics, and Aggregate Outcomes

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Introduction

We will cover recent research exploring the relation between firm-level frictions, firm dynamics, and their aggregate implications.

The main objectives of the course are the following:

1) Illustrate empirical methods used to identify firm-level frictions, and quantify their aggregate implications.

2) Derive and solve a basic firm dynamics framework with heterogeneous firms.

3) Extend the basic model and introduce frictions. Although we will mostly analyze financial factors, we will illustrate a general framework applicable to study also the effects of other frictions.

4) Applications: we will review selected recent research on the aggregate implications of financial frictions and firm dynamics.

Content

1. **Introduction to Firm Dynamics and Misallocation**
    a. Stylized facts on firm dynamics.
    b. Dispersion in productivity, frictions, and misallocation of resources: theoretical framework and empirical evidence.
2. **Financial frictions and Firm Dynamics: empirical evidence and theory**
    a. Micro foundation and empirical testing of firm level financial frictions, and of their aggregate implications.
    b. Firm dynamics and financial frictions: basic model and applications
        i. Uncertainty, bankruptcy risk, and selection.
        ii. Radical and Incremental innovation.
3. **Finance, firm dynamics and the business cycle: theory and applications**
    a. Credit Cycles, with and without banks. Empirical Evidence and Theory.

Syllabus

1) Introduction to Firm Dynamics and Misallocation.

Papers de noted with a * will be discussed in the lectures

- *John Asker, Allan Collard-Wexler, and Jan De Loecker, Dynamic Inputs and Resource (Mis)Allocation, Journal of Political Economy 2014 122:5, 1013-1063
- *Mark Bils, Peter J. Klenow, Cian Ruane, Misallocation or Mismeasurement?, Stanford University, December 4, 2018
- Lucia Foster, Cheryl Grin, John C. Haltiwanger, Zoltan Wolf, 2018, Innovation, Productivity Dispersion and Productivity Growth. NBER Working Paper 24420
- *Hsieh, C. and P.J. Klenow (2009). "Misallocation and Manufacturing TFP in China and India", The Quarterly Journal of Economics 124, 1403-1448.
- *Joel David, Venky Venkateswaran, 2019, The Sources of Capital Misallocation, American Economic Review, forthcoming.
- *Haltiwanger, John, Robert Kulick and Chad Syverson. 2018. "Misallocation Measures: The Distortion that Ate the Residual." NBER Working Paper No. 24199
- *John Haltiwanger, Ron S. Jarmin, Robert Kulick, Javier Miranda, High Growth Young Firms: Contribution to Job, Output, and Productivity Growth, Chapter in NBER book Measuring Entrepreneurial Businesses: Current Knowledge and Challenges (2017), John Haltiwanger, Erik Hurst, Javier Miranda, and Antoinette Schoar, editors (p. 11 - 62).
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2) Financial frictions and Firm Dynamics: empirical evidence and theory

Empirical evidence

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- Santiago Bazdresch, R. Jay Kahn, Toni M. Whited, Estimating and Testing Dynamic Corporate Finance Models, The Review of Financial Studies, Volume 31, Issue 1, January 2018, Pages 322–361
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- *Chodorow-Reich, G. (2014). The Employment Effects of Credit Market Disruptions: Firm-level Evidence from the 2008–9 Financial Crisis. The Quarterly Journal of Economics, 129(1), 1-59.

- Dell'Ariccia, Giovanni, Detragiache, Enrica, and Rajan, Raghuram, 2008, "The real effect of banking crises," Journal of Financial Intermediation, Elsevier, vol. 17(1), pages 89-112.
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- Hadlock, Charles J., and Joshua R. Pierce, 2010, New evidence on measuring financial constraints: Moving beyond the KZ Index, Review of Financial Studies 23, 1909–1940.
- Kaplan, Steven N., and Luigi Zingales, 1997, Do investment-cash flow sensitivities provide useful measures of financing constraints?, Quarterly Journal of Economics 115, 707–712.
- Simon Gilchrist, Jae Sim and Egon Zakrajsek, 2013, ``Misallocation and Financial Frictions: Some Direct Evidence from the Dispersion in Borrowing Costs", Review of Economic Dynamcis, January 2013.
- *Nikolov, Boris and Schmid, Lukas and Steri, Roberto, The Sources of Financing Constraints (November 30, 2018). Swiss Finance Institute Research Paper No. 18-74. Available at SSRN: https://ssrn.com/abstract=3293849 or http://dx.doi.org/10.2139/ssrn.3293849

Theory

- *Hopenhayn, Hugo A, 1992. "Entry, Exit, and Firm Dynamics in Long Run Equilibrium," Econometrica, Econometric Society, vol. 60(5), pages 1127-1150, September.
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- *Caggese, A., and V. Cuñat, 2013, "Financing Constraints, Firm Dynamics, Export Decisions, and Aggregate Productivity", Review of Economic Dynamics, Special Issue on Misallocation and Productivity, edited by Diego Restuccia & Richard Rogerson, vol. 16(1), pages 177-193, January 2013.
- *Caggese, A., 2019, "Financing Constraints, Radical versus Incremental Innovation, and Aggregate productivity", American Economic Journal: Macroeconomics.
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- *Midrigan, Virgiliu, and Daniel Yi Xu. 2014. "Finance and Misallocation: Evidence from Plant-Level Data." American Economic Review, 104(2): 422-58
- Moll, Benjamin. 2014. "Productivity Losses from Financial Frictions: Can Self-Financing Undo Capital Misallocation?" American Economic Review, 104(10): 3186-3221.
- Oberfield, Ezra, 2013. Productivity and misallocation during a crisis: Evidence from the Chilean crisis of 1982. Review of Economic Dynamics 16 (1), 100–119
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- Acemoglu, D., U. Akcigit, N. Bloom, and W. R. Kerr (2013). Innovation, Reallocation and Growth. National Bureau of Economic Research WP 18993.
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- Daniel Garcia-Macia, Chang-Tai Hsieh, Peter J. Klenow, 2019, How Destructive is Innovation?, working paper.
- Pugsley, Benjamin, Petr Sedlacek, and Vincent Sterk, "The Nature of Firm Growth," CEPR Discussion Papers 12670, C.E.P.R. Discussion Papers January 2018.
- Sedlacek, P., and V. Sterk, 2016, The Growth Potential of Startups over the Business Cycle, American Economic Review.

3) Finance, firm dynamics and the business cycle: theory and applications

- *Arellano, C., Yan, Bai, and Patrick Kehoe, 2019, Financial Frictions and Fluctuations in Volatility, Journal of Political Economy, Forthcoming.
- Bernanke, B. S., M. Gertler, and S. Gilchrist, "The Financial Accelerators in a Quantitative Business Cycle Framework," in John B. Taylor and Michael Woodford, eds., Handbook of Macroeconomics, 1999, pp.1341-1393.
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